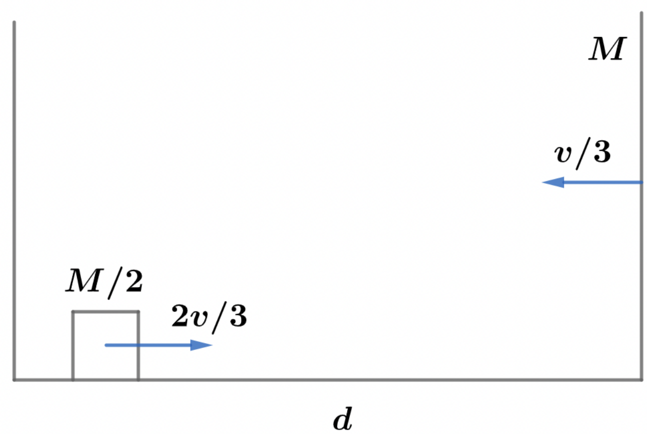
2022B F=ma Exam: Problem 17

Kevin S. Huang

The velocity of the center of mass is

$$v_{\text{CM}} = \frac{(M/2)(2v) + Mv}{M + M/2} = \frac{2Mv}{3M/2} = \frac{4v}{3}$$



In the CM frame, it takes time

$$T = \frac{d}{\frac{2v}{3} + \frac{v}{3}} = \frac{d}{v}$$

for the puck to hit the box. Recall the velocities of the objects flip after an elastic collision in the CM frame. Thus, the relative speed is still v and it takes the same time T to the next collision. The answer is $\boxed{\text{B}}$.